CET257 Enterprise Project — Group Meeting Minutes: Week 3

Faculty: Faculty of Technology, School of Computer Science
Module: CET257 Enterprise Project
Module Leader: Dr Becky Allen
Supervisory Tutor: Rupak Chaudhary
Team Number: 9 | **Team Name:** SARAS
Meeting Reference: MIN-WK03-20260805
Meeting Date: Wednesday, 5 August 2026
Meeting Time: 08:35:05 – 09:17:01 NPT (Duration: 41 minutes 56 seconds)
Location / Platform: Microsoft Teams (Meetings Channel)
Attendees: Rijan Kapur Poudel (PM/Backend), Sushank Jung Rawal (Deputy PM/UI Lead)

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1. Attendees and Day's Activities

Attendance Register

- **Rijan Kapur Poudel** — Present (Joined 08:35:05, Left 09:17:01)
- **Sushank Jung Rawal** — Present (Joined 08:35:18, Left 09:16:55)

Day's Activities and Agenda

1. **Pre-Consultation Rehearsal:** Conducting a structured walkthrough of the presentation plan for tomorrow's formal client consultancy session with Mr. Ninson Lama (scheduled for Thursday, 6 August at 14:00 NPT).
2. **Review of Five Selected Inquiry Themes:** Finalizing the exact wording of the five questions covering learner profiles, fire safety priorities, hardware environments, explanatory feedback, and training compliance boundaries.
3. **Database Concurrency Bug Verification:** Reviewing and testing PostgreSQL Migration 010 (Commit `5da69c7`), which introduced composite unique constraints on `user_progress (user_id, module_id, lesson_id)` to resolve race conditions.
4. **Interactive Question Formats:** Evaluating API endpoint logic for multiple question formats (single-choice, multi-select, sequencing) in `apps/web/src/lib/quiz-data.ts`.
5. **System Error Boundaries:** Implementing React Error Boundary wrappers to ensure graceful failure recovery if external assets fail to load.

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2. Breakdown of Achieved Activity for the Week

Rijan Kapur Poudel

- Authored and dispatched the formal consultation invitation to client Mr. Ninson Lama (03 August, 10:00 NPT); received confirmation on 04 August at 09:20 NPT.

- Authored and deployed Supabase Migration 010, successfully implementing composite uniqueness rules and `ON CONFLICT DO UPDATE` upsert logic, completely eliminating the progress persistence race condition.
- Restructured API route handlers to serve rich question schemas, including mandatory fields for detailed post-answer explanatory feedback.
- Implemented React Error Boundary components around data-fetching hooks to prevent white-screen crashes on network dropouts.
- Formulated the formal question sheet and meeting protocol for the Week 3 client session.

Sushank Jung Rawal

- Successfully integrated the Photo Sphere Viewer prototype into a dedicated Next.js client component, verifying smooth 360° rotational rendering.
- Configured preliminary hotspot coordinates and dynamic tooltip markers on the equirectangular panorama.
- Prepared visual slides and prototype screen captures for the 20-minute client consultancy session.
- Finalized `03 System Requirements Analysis.pdf` (Task A1), completing structured flowcharts for learner authentication, lesson progression, and quiz scoring.
- Audited mobile viewport rendering down to 375px width to ensure navigation controls do not clip on smaller screens.

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3. Breakdown of Planned Activity for the Week Ahead (Week 4)

Rijan Kapur Poudel

- Lead the technical discussion during the client consultation on 6 August; draft and send the post-consultation written confirmation on 7 August.
- Transition assessment scoring logic entirely to server-side PostgreSQL stored procedures (RPC) to prevent client-side answer key inspection in browser DevTools.
- Implement database Row-Level Security (RLS) policies on user data tables.
- Refactor Task A5 project schedule based on client directives received during the consultation.

Sushank Jung Rawal

- Present the UI/UX architecture and learner journey during the client consultation on 6 August.
- Audit and overhaul keyboard accessibility across the lesson wizard: replace drag-and-drop sequencing with accessible "Move Up" / "Move Down" button controls (WCAG 2.5.7).
- Implement `prefers-reduced-motion` CSS media query handlers to support users sensitive to rotational animations.
- Begin authoring the formal corporate identity manual for Task A6 (`SARAS Labs Corporate Identity.docx`).

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4. Project Progress This Week

- **Client Consultation Prepared:** Complete 20-minute session structure, slides, and five inquiry themes validated and ready for delivery.
- **Critical Bug Resolved:** Progress-save race condition permanently eliminated via database constraint refactoring (Migration 010).
- **360° Panoramic Viewer Operational:** Prototype canvas renders equirectangular environments smoothly in standard web browsers.
- **Task A1 Milestone Complete:** All three core design documents (Requirements Spec, User Analysis, System Analysis) finalized and ready for integration.

5. Issues Encountered and Planned Response

Issue Description	Severity	Impact	Planned Team Response
Client-Side Answer Leakage: Initial quiz implementation included answer keys in the client JSON bundle, allowing technically adept users to inspect correct answers via browser DevTools.	High	Compromises assessment integrity and invalidates safety certification.	**Response:** Rijan will shift scoring logic to a server-side PostgreSQL function (RPC), stripping correct answer flags from client payloads.
Accessibility Barrier in Drag-and-Drop: Step reordering in hazard sequencing required single-pointer dragging, which fails WCAG 2.2 Level AA accessibility for keyboard-only and motor-impaired learners.	High	Fails university accessibility criteria (WCAG 2.5.7).	**Response:** Sushank will implement accessible button-driven reorder controls alongside drag handles.

6. Formal Decisions and Action Items

Decisions

1. **D-07:** Implement all assessment score calculations on the server via PostgreSQL RPC functions.
2. **D-08:** Strictly adhere to the client consultation follow-up protocol: send formal written summary within 24 hours and await written sign-off before modifying requirements.
3. **D-09:** Standardize all interactive feedback components to display both root-cause hazard explanations and immediate corrective actions.

Action Items

Action Item	Assigned Lead	Target Date	Status
Deliver 20-minute client consultation with Mr. Ninson Lama	Rijan & Sushank	06 Aug 2026	Scheduled
Send formal written summary email to Mr. Lama	Rijan Kapur Poudel	07 Aug 2026	Scheduled
Implement accessible 'Move Up' / 'Move Down' reorder buttons	Sushank Jung Rawal	10 Aug 2026	In Progress

Action Item	Assigned Lead	Target Date	Status
Author server-side assessment scoring RPC function	Rijan Kapur Poudel	12 Aug 2026	In Progress

Minutes submitted to Supervisory Tutor: Rupak Chaudhary | Recorded by Rijan Kapur Poudel | Approved by Sushank Jung Rawal